41014 Sensors and Control in Mechatronics Systems

Assessment 4: Integrated Project

Marking Guide

Weighting	40% of the subject
Due	Exam Week
Submission	Canvas (links to GitHub project page and video), followed by individual viva voce
Group size	3–4 members for students enrolled in 41014; 1–2 members for students enrolled in 42043
Platform	Simulation component compulsory for all students; real hardware implementation optional (see Section 1)
Format	Team submission: working pipeline, GitHub project page with code, and a 5-minute demo video. Individual viva voce for each team member.

1 Task summary

This is a group project. Groups have 3–4 members for students enrolled in 41014, and 1–2 members for students enrolled in 42043. Teams integrate sensing, estimation and control on a simulated platform. Two tracks are offered and may be personalised, or a team may propose its own project of equivalent scope.

Simulation and hardware

A simulation component is compulsory for all students, and all five criteria are assessed on it. Real hardware implementation is available to teams who have demonstrated their simulation component working by the end of Week 11, which is required for the compulsory hardware induction. Teams who have not demonstrated a working simulation by the end of Week 11 cannot attend the induction and complete the project in simulation only.

Hardware work is optional and is not required for any band up to Distinction. A working hardware implementation of the team's pipeline, shown in the video and reported on the project page, is marked as High Distinction evidence: it satisfies the independent contribution expected at that band for Part B (sensing and calibration on the real sensor), Part D (control on the real platform) and Part E (robustness under real noise and disturbance), provided the work is also of Distinction standard in that criterion. A quantitative comparison between simulation and hardware results strengthens the case. A hardware result does not replace missing simulation evidence, and a hardware attempt that does not complete the task is marked on what it shows.

Track A, mobile robot: visual waypoint navigation. TurtleBot3 with 2D LiDAR and camera in MuJoCo. Navigate a room with obstacles using the LiDAR and visually dock to the ArUco-marked charging station. Required: state-space model of the platform; EKF fusing wheel-encoder odometry with range and bearing, or pose, measurements from the camera; state-feedback or LQR trajectory tracker. Constraint: the controller must use the estimated state, never ground truth.

Track B, manipulator: vision-guided pick. UR3 in MuJoCo with a wrist-mounted RGB-D camera. Detect a target object on a tabletop, plan an approach and execute a pick using IBVS or PBVS. Required: camera calibration; depth-based object localisation; state-space joint controller;

observer for unmeasured states. Constraint: object position is randomised between trials and the system must demonstrate robustness.

Own project. Must include, at minimum, a state-space model with controllability and observability analysis, at least one calibrated exteroceptive sensor, a Kalman filter or observer, and a feedback controller running on estimated state. Scope should be agreed with the teaching team before Week 10.

Part	Criterion	Marks
A	Modelling: correctness of the state-space model, observability and controllability analysis	20
B	Sensing: calibration, noise handling and frame management	20
C	Estimation: KF/EKF or observer design, tuning and reported performance	20
D	Control: controller design, tuning and tracking results	20
E	Integration and robustness: performance under disturbance and noise	20
Total (team mark)		**100**

The team mark is assessed from the project page, the code and the demo video. Each student's individual mark is the team mark moderated by their viva voce (Section 4).

2 Grade bands

Band descriptors are cumulative. To reach a band, a submission must satisfy the descriptors beneath it.

Band	Range	Descriptor
Fail	0–49	The pipeline does not run end to end, or one or more of the required components (model, sensing, estimator, controller) is missing. The controller uses ground truth. Results are absent, or shown without evidence they come from the submitted code. The video or project page is missing or does not show the system working.
Pass	50–64	Satisfactory work meeting the basic requirements. The team adapts material from the weekly tutorials to the chosen platform. All required components are present and the task is completed in at least the nominal case. Results are shown but described rather than explained. Tuning values are stated without justification.
Credit	65–74	Correct and complete pipeline that goes beyond reuse of tutorial code. The model is derived for the actual platform and its assumptions are stated. Noise parameters and gains are chosen deliberately and defended. Results are quantitative (error metrics, success rates) rather than only visual, and the discussion explains why the system behaves as it does.

Band	Range	Descriptor
Distinction	75–84	As above, plus systematic evaluation. Performance is reported over repeated trials with varied conditions, and failures are diagnosed and traced to specific causes in the model, sensing, estimator or controller. Trade-offs are quantified, for example estimator consistency against responsiveness, or control effort against tracking error. The project page and code are clean, reproducible and would be usable by another team.
High Distinction	85–100	As above, plus independent contribution. Evidence of engagement past the subject material including a working implementation on real hardware (Section 1), for example a justified extension of the method, an additional sensor or estimator compared rigorously, a robustness study the brief did not ask for, or a well-argued account of the limits of the approach. Analysis is rigorous, quantitative and honest about what the results do and do not support.

Note on tutorial material. Adapting code from the weekly tutorials is expected and is not plagiarism. It is the intended starting point, and doing it competently on the project platform is what a Pass looks like. Marks above a Pass come from what the team adds: models derived for their platform, defended design choices, quantitative evaluation, and analysis that explains the results they produced.

3 Deliverables checklist

Markers should confirm the following before assigning bands. A missing deliverable caps the affected criteria at Pass unless the evidence is available elsewhere in the submission.

- **Project page (GitHub).** States the track or own project, the model, the frame diagram, estimator and controller design with parameter values, and quantitative results. Install and run instructions are present and the code runs as described.
- **Code.** Organised, readable, and runs from a fresh environment. Ground truth from the simulator is used only for evaluation, and this is visible in the code.
- **Demo video (5 minutes).** Shows the system completing the task, at least one varied or disturbed trial, and the estimate and its uncertainty alongside the true state. Content beyond 5 minutes need not be considered.
- **Contribution statement.** Each member's contribution is recorded on the project page.
- **Generative AI declaration.** Tools and purpose stated on the project page.

4 Individual viva voce

Each student attends a short viva in which they are asked about the team's video and project page. Questions should cover the student's own contribution and at least one component they did not lead, so that every student can explain the pipeline as a whole. Suggested prompts:

- Point to a result on the project page and explain why it looks the way it does.

- Which states are observable from which sensors, and what happens when the marker or object leaves the view?
- Where do the values in Q and R come from, and what would happen if R were ten times smaller?
- How were the controller gains or LQR weights chosen, and what limits them?
- Walk through the frames from the sensor measurement to the quantity the controller acts on.
- What failed during development, how was it diagnosed, and what would you change?

The viva moderates the team mark into an individual mark:

$$\text{Individual mark} = \text{Team mark} \times \text{Viva factor}.$$

Viva outcome	Factor	Descriptor
Full understanding	1.0	Explains their own contribution and the rest of the pipeline accurately, links results to design choices, and answers follow-up questions with reasons.
Partial understanding	0.8	Explains their own contribution well but is unsure of other components, or describes what the system does without explaining why.
Limited understanding	0.5	Can describe parts of their own contribution only in general terms, and cannot explain key results or design choices on the project page.
No demonstrated understanding	0.0	Unable to explain the submission, or did not attend without approved special consideration. Refer to the academic integrity process where appropriate.

5 Academic integrity

Teams may discuss concepts with other teams but must submit their own code, project page and video. Generative AI use must be declared on the project page, stating tools and purpose. Non-declaration is itself a matter for the integrity process.

Code adapted from the weekly tutorials, MuJoCo examples and the Robotics Toolbox is expected and should be acknowledged. Distinguish shared structure inherited from these sources, which is expected, from identical code, figures or text shared between teams, which is not.